1 Introduction

As the Introduction explains, there are errors associated with using parameterised models to calculate free energies of binding. Because of this, studies are starting to appear that calculate the quantum corrected binding free energies. These studies use the models provided by classical mechanics as guiding potentials, when combining these models with molecular dynamics (MD), structures can be generated computationally cheaply. Once the energies of these structures have been calculated, both at the reference potential and desired potential, a single step perturbation, such as the Zwanzig equation, can be performed. Due to the large size differences between these energies, interaction energies are commonly used. However, the Zwanzig equation has not been derived with the intention of using interaction energies so this is an approximation. Furthermore the use of interaction energies makes the approximation that the conformations sampled by a protein when a ligand is bound, will be sampled if the protein is unbound.

Within this chapter the assumption that these structures represent quantum structures is made, and by using the extended thermodynamic cycle, the Zwanzig equation is applied to obtain the quantum corrected free energies.

THE INTRODUCTION WILL EXPLAIN WHY ACCURATE FREE ENERGIES ARE REQUIRED - BUILD UPON THIS HERE AND TALK ABOUT APPLYING QUANTUM CORRECTIONS?

2 Theory

DERIVE CUMULANT EXPANSION TO FREE ENERGY

BERG FORMULA

3 Methods

EQUILIBRATION

APPLICATION OF THE EXTENDED THERMODYNAMIC CYCLE TO STRUCTURES

ONETEP DETAILS

CHARGE PERTURBATION

4 Results

4.1 Applying a Single Step Perturbation Approach to Non-Polar and Polar Test Systems

To validate the method simple test systems were selected, these included ethane in 133 water and ethanol in 138 water. These were selected to show how the obtained results are affected by polar and non-polar systems. Initially, the quantum simulations were performed with NGWFs as detailed in section 3. In order to obtain classical structures a 40 ns MD simulation was performed, the resulting structures were divided into two groups, configurations that were generated in the first 20 ns (state A) and configurations generated in the second 20 ns (state B). 400 snapshots were then extracted from each state and the results compares in table 1.

Examination of the structures that did not converge due to SCF convergence errors yielded no explanations. The convergence issues came from the outer loop of ONETEP, the NGWF optimisation !!!REF ME!!!!!!.

Table 1: MM to QM perturbation applied to ethane using NGWFs, allowing upto 10% of snapshots to be omitted due to SCF convergence errors. All energies shown are in kcal/mol

Number of snapshots	1^{st} 20.0 ns Δ G	2^{nd} 20.0 ns Δ G	Difference $\Delta\Delta$ G
400	-1447906.57	-1447912.01	5.44

It is apparent that 400 snapshots from state A and B are not enough to provide convergence of the method. As such additional snapshots were calculated, but for speed, these snapshots were calculated using PAOs instead of NGWFs. Simulation times for the single point energy calculations when using NGWFs were approximately 7 hours on 6 cores, whereas by using PAOs, this computational time lowered to 4 hours on 6 cores. Additionally, by using PAOs the previous convergence issues were not present as only the density kernel is optimised !!!REF ME!!!!!!. By increasing the number of snapshots from each state to 1000 and applying this SSP method, $\Delta\Delta G$ is still not 0. These results can be seen in table 2.

Table 2: MM to QM perturbation applied to ethane using PAOs. All energies shown are in kcal/mol

Number of snapshots	1^{st} 20.0 ns Δ G	2^{nd} 20.0 ns Δ G	Difference $\Delta\Delta$ G
1000	-1442495.51	-1442499.82	4.31

Investigating the lack of convergence present within the $\Delta\Delta$ G led to calculating the free energy as a function of the number of snapshots included. In this way if the free energy is being dominated by one or a few snapshots it will become apparent. The resulting graph can be seen in figure 1.

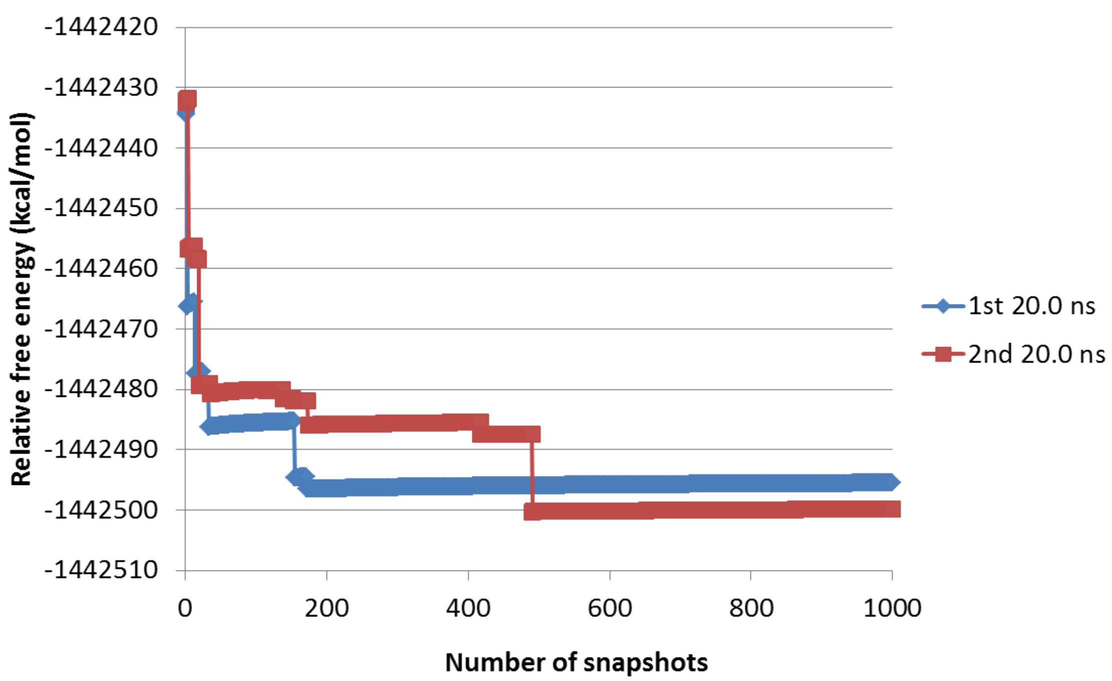

Figure 1: Free energy as a function of the number of snapshots for ethane in 133 water, all values shown are in kcal/mol

The "sawtooth" shape of the graph in figure 1 is characteristic of poor sampling !!!REF ME!!!and according to the equation XXX will take XXX snapshots to converge.

In order to ensure that the lack of convergence in the $\Delta\Delta$ G was related to using a basis set with poor accuracy, when applying the SSP method to ethanol in 138 water, NGWFs were used. Additionally the original MD simulation used to generate configurations was run for a longer time of 200ns to provide certainty of obtaining uncorrelated structures. However, regardless of these additional measures taken to achieve convergence of $\Delta\Delta$ G, a higher number can be seen (table 3) and 10% of structures would not converge; an issue that, again, was not explained by further examination of these structures.

Table 3: MM to QM perturbation applied to ethanol using NGWFs, allowing upto 10% of snapshots to be omitted due to SCF convergence errors. All energies shown are in kcal/mol

Number of snapshots	1^{st} 20.0 ns Δ G	2^{nd} 20.0 ns Δ G	Difference $\Delta\Delta$ G
1000	-1511726.64	-1511765.45	38.81

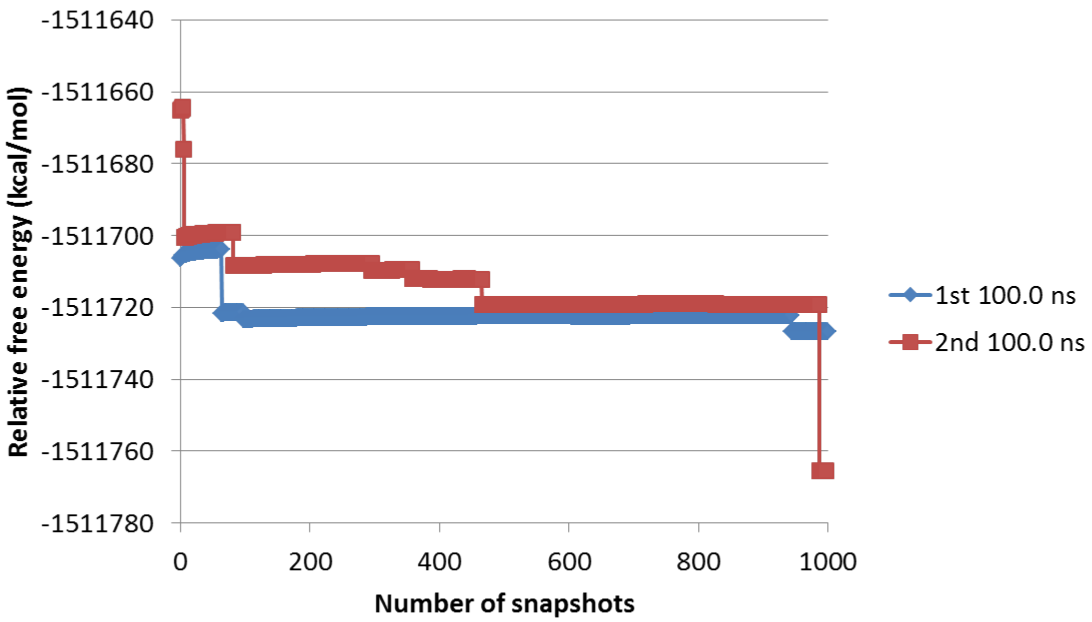

Figure 2: Free energy as a function of the number of snapshots for ethanol in 138 water, all values shown are in kcal/mol

A similar graph as produced for ethane in 133 water, was then produced for ethanol in 138 water and shown in figure 2. This figure shows the catastrophic effect that one of these outlier structures can have on the free energy. Applying equation XXX to this system shows that XXX snapshots are required for convergence. The largest affect to the free energy occurred within this 2nd 100 ns simulation of ethanol in 138 water. As such, the distributions of energies produced by the structures within this simulation were compared.

Three energy distributions were produced from this simulation, these energies were the MM, QM and QM-MM distributions. To analyse these distributions, they were plotted within box and whisker plots. The “box” in the box and whisker plots shows the interquartile range (IQR) of the data, and the line in the middle of the box is the median. This whisker length is 1.5 times the length of the IQR, any snapshots outside of this length can be thought of as an energetical outlier.

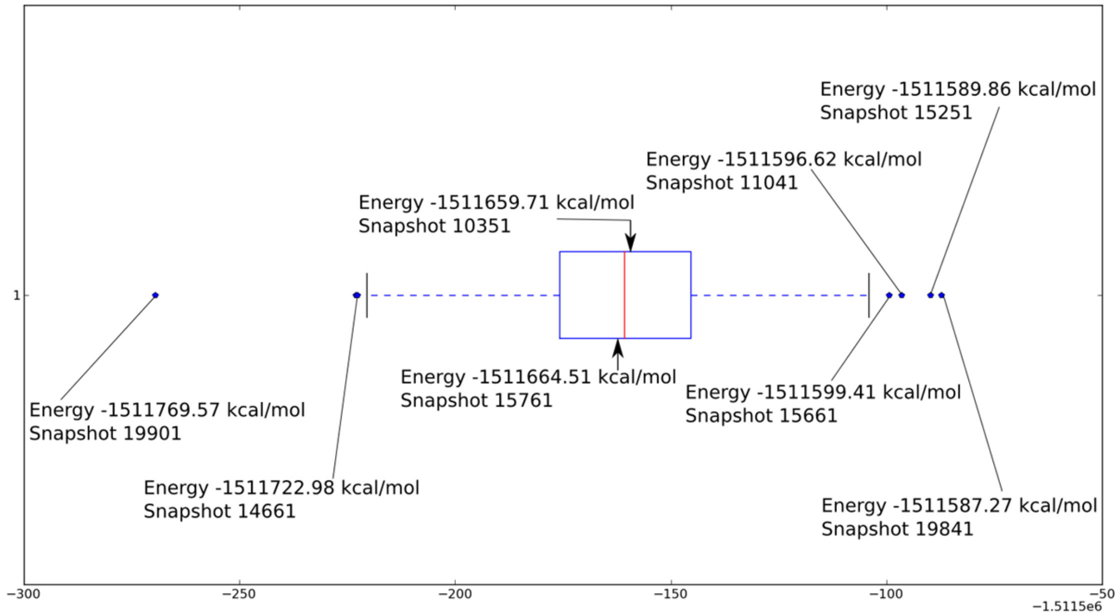

Figure 3: Energy difference distribution for the ethanol in 138 water, 2nd 200 ns, plotted in a box and whisker plot

Six outlier snapshots can be identified from figure 3 and, for analysis purposes, two snapshots close to the median value were selected. Another interesting piece of information that can be easily seen from the above box and whisker plot is the large range of energy differences of 182.30 kcal/mol. Producing the same plots for the MM and QM energy distributions provides some valuable results.

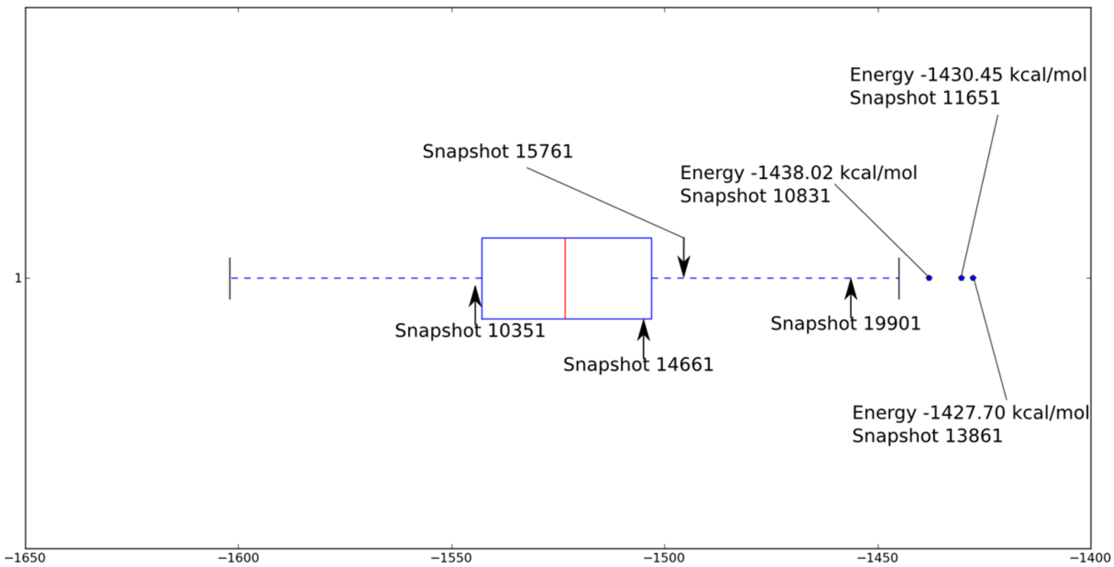

Figure 4: MM energy distribution for the ethanol in 138 water, 2nd 200 ns, plotted in a box and whisker plot

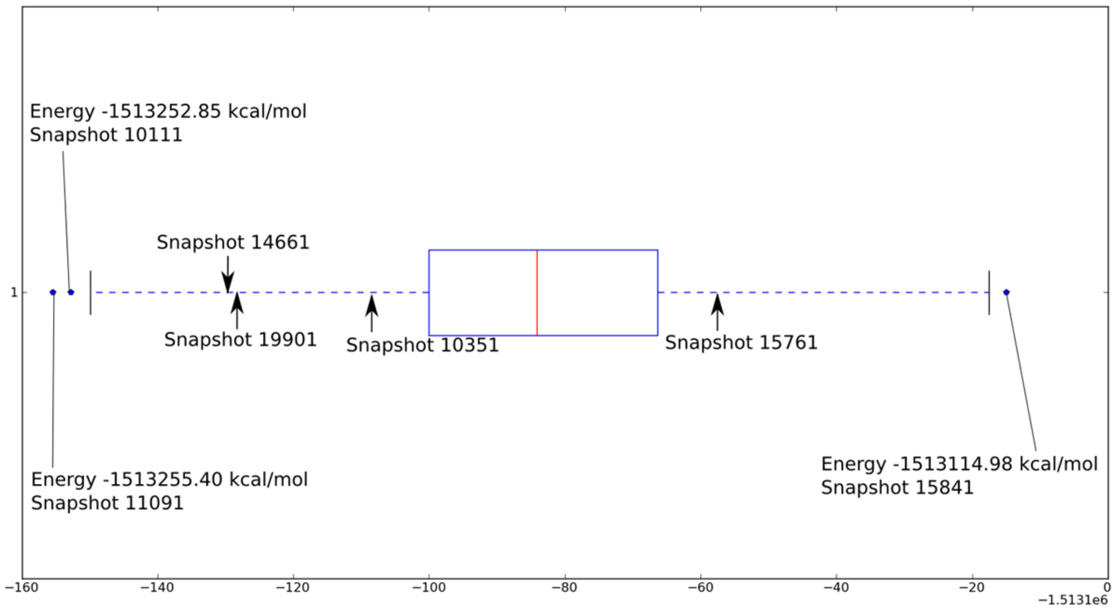

Figure 5: QM energy distribution for the ethanol in 138 water, 2nd 200 ns, plotted in a box and whisker plot

Comparing which snapshots were classed as outlier snapshots in each distribution produced the following table (table 4).

Table 4: Outlier snapshots

Energy Difference	QM	MM
19901	11901	10831
14661	10111	11651
15661	15841	13861
11041		
15251		
19841		

Table 4 shows that for all three distribution, there are no common outliers. Which proves that the large jumps present in figure 3 attributed to the energy difference outliers are not present due a structure not being described well in either the MM or QM distributions. Additionally, the examination of the positions of the snapshots identified as close to the median in the energy differences within the MM and QM distributions show that, in both cases, the snapshots are outside of the IQR. However, because both of the snapshots appear on the same side of the MM and QM distributions i.e. either both snapshots appear on the more negative side of the IQR or both snapshots appear on the less negative side of the IQR, the snapshots appear close to the median in the energy difference distribution. With regard to the outlier snapshots within the energy difference distribution, the more negative side of the distribution will have a larger effect on the free energy. This is due to the calculation of the exponential of the negative energy present within the Zwanzig equation (equation XXX !!!REF ME!!!). Finding these snapshots within the MM and QM distributions show them to be on different sides of the distributions. This difference shows that the MM and QM configurational space does not overlap exactly, and this inexact overlap of configurational space causes these energy difference outliers.

4.2 Controlling the Level of Configurational Overlap to Measure the Convergence of the Free Energy

In order to investigate the perturbation of one potential to another potential while controlling the degree of configurational space overlap, a test system was set up. Calculating quantum energies are orders of magnitude more computationally expensive than classical energies, as such, the test system will be between classical systems. The test system of ethane in 133 water was used with the setup described within the methods section. The MD simulation was then divided into two parts and 10000 snapshots were extracted at regular intervals and the charges present on ethane were then perturbed. The standard charges on ethane used were obtained from antechamber [1], these were ...... The perturbations then involved changing these charges and calculating the new energies, these charges were changed by doubling them, tripling them etc.. up to eight times the charge. The convergence of ΔΔ G was then measured while lowering the configurational space overlap.

Table 5: Calculating the free energy required to perturb the charges on ethane in 133 water from standard charges to increased charges. Sampling performed using standard charges. All energies shown are in kcal/mol

Increased charge	1^{st} 100.0 ns Δ G	2^{nd} 100.0 ns Δ G	Difference ΔΔ G
Double	2.800	2.798	0.002
Triple	7.439	7.434	0.005
Quadruple	13.898	13.886	0.012
Pentuple	22.148	22.111	0.037
Sextuple	32.231	32.108	0.123
Septuple	43.957	43.635	0.322
Octuple	57.365	56.751	1.206

Table 5 shows that by lowering the amount of overlap present between the two states involved in the perturbation, the free energy does not converge. The trend of ΔΔ G is as expected, when perturbations are small the free energy converges. Configurational space overlap is demonstrated in figure 6 and 7. Where, in figure 6, the carbon carbon bond length is measured when an MD simulation is performed using the altered charges. Figure 7 shows a similar analysis, but using the hydrogen-carbon-carbon angle of these MD simulations. These figures show that as the charge is increased, both the bond length and angle is increased slightly.

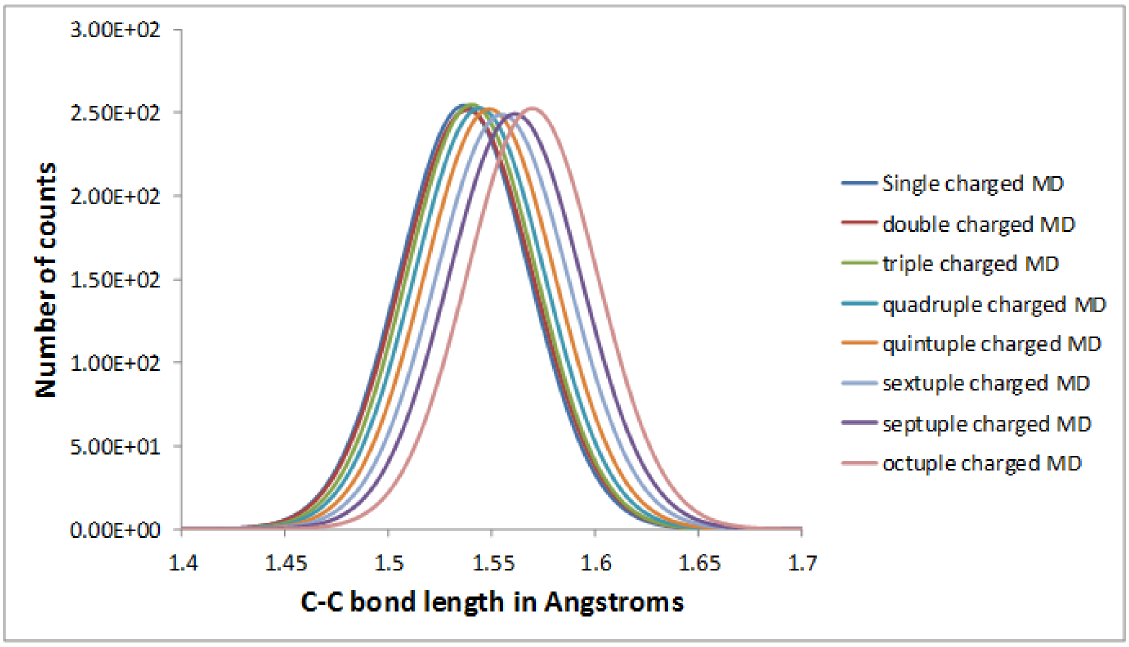

Figure 6: Carbon-carbon bond length within 8 different MD simulations with different partial charges on ethane

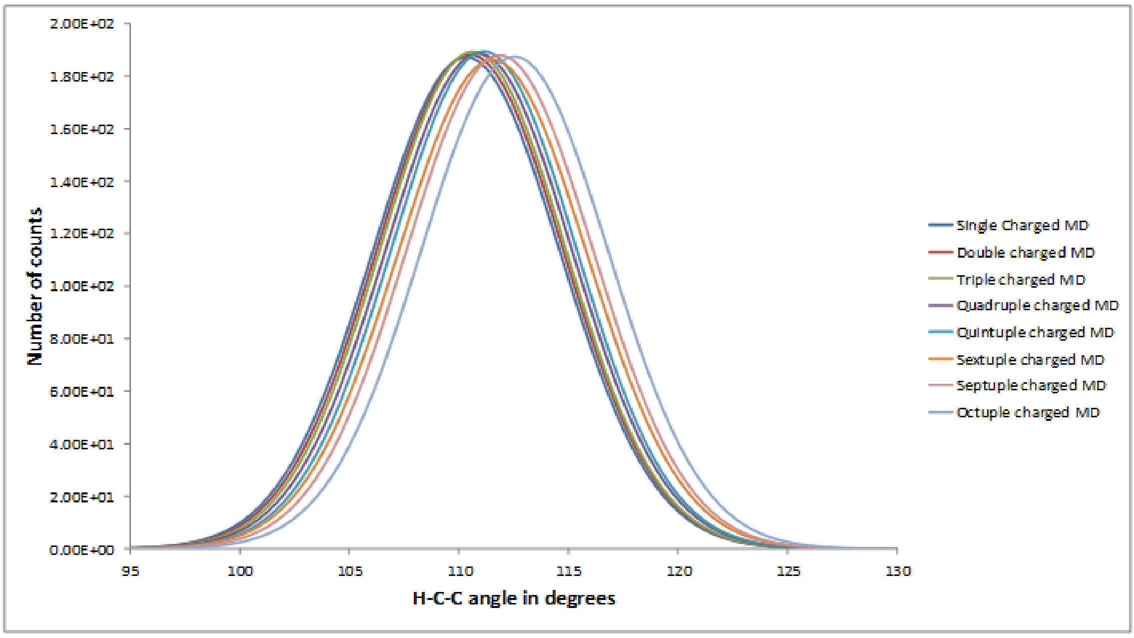

Figure 7: Carbon-carbon-hydrogen angle within 8 different MD simulations with different partial charges on ethane

Noticing how poor the configurational space overlap is between the standard and octuple charged systems, the reverse calculation was performed. This involved running an MD simulation at the desired charge and perturbing to a standard charged system. Free energy is a state function, so the reverse process should be the negative of the forward process.

Table 6: Calculating the free energy required to perturb the charges on ethane in 133 water, from standard charges to an increased charge. Sampling performed using the increased charges. All energies shown are in kcal/mol

Increased charge	1^{st} 100.0 ns Δ G	2^{nd} 100.0 ns Δ G	Difference $\Delta\Delta$ G
Double	2.773	2.771	0.002
Triple	7.290	7.293	-0.003
Quadruple	13.386	13.428	-0.042
Pentuple	20.864	20.795	0.069
Sextuple	29.422	29.228	0.194
Septuple	36.360	37.766	-1.406
Octuple	43.939	45.746	-1.807

Table 7: The difference between perturbations, each value is the average of the 1^{st} and 2^{nd} 100ns. All energies shown are in kcal/mol

Increased charge	standard to increased charge Δ G	increased charge to standard Δ G	Difference $\Delta\Delta$ G
Double	2.799	2.772	0.027
Triple	7.437	7.292	0.145
Quadruple	13.892	13.407	0.485
Pentuple	22.130	20.830	1.300
Sextuple	32.170	29.325	2.845
Septuple	43.796	37.063	6.733
Octuple	57.058	44.843	12.215

The results show poor convergence of the free energy. This shows that when the configurational space does not overlap exactly the forward and reverse process does not converge. The lack of overlap here is small, the lack of overlap between the classical and quantum is very poor, this offers some explanation toward the poor convergence of the MM to QM. It is clear that without sampling a large amount of configurational space, the free energy from MM to QM will not converge. Also 10000 snapshots for ethane in 133 water is a manageable computational cost, but for a larger, more biologically relevant system, 10000 snapshots will be extremely computationally expensive. So several post-processing methods were applied to try and converge these free energies and the results are shown in the next section.

4.3 Methods to Converge the Free Energy

The largest affect an energy difference outlier had on the convergence of free energies was observed for ethanol in 138 water in the 2^{nd} 100 ns. Because of this, all attempts to converge the free energy, using the data already obtained, will be for this system. Section !!!REF ME!!!shows that it is impossible to predict where energy difference outliers will occur by examining only the classical or quantum energies. Because of this, the first attempt to better converge the free energy was to subtract the average energy from both energy distibutions,

$$\Delta E = (E_i^{MM} - \langle E^{MM} \rangle) - (E_i^{QM} - \langle E^{QM} \rangle). \tag{1}$$

Where the subscript i represents the i^{th} structure, $\langle E_i^{MM} \rangle$ is the average classical energy.

The results, shown in table 8, display the same poor convergence of table 3. Applying the Zwanzig equation [2] as a function of the number of snapshots, provides large "jumps" in free energy again (figure 8), showing that subtracting the average from each energy distribution cannot converge the free energy.

Table 8: MM to QM perturbation applied to ethanol using NGWFs, each energy distribution has the average energy subtracted from it. Allowing upto 10% of snapshots to be omitted due to SCF convergence errors. All energies shown are in kcal/mol

Number of snapshots	1^{st} 20.0 ns Δ G	2^{nd} 20.0 ns Δ G	Difference $\Delta\Delta$ G
1000	-66.04	-104.51	38.47

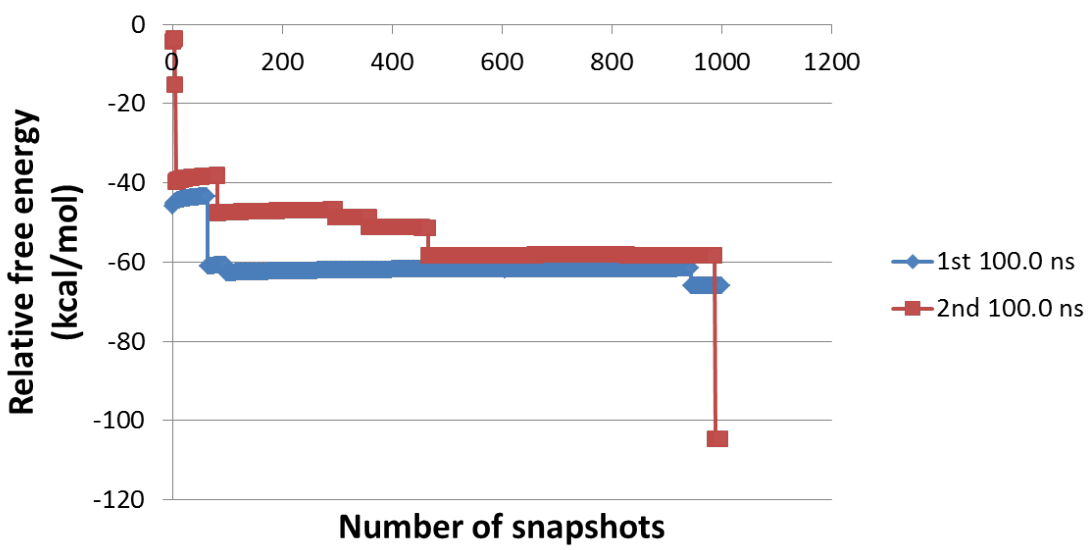

Figure 8: Free energy as a function of the number of snapshots included for ethanol in 138 water, subtracting the average energy from energy distributions

Following this, the distribution of the energy differences was examined and found to be Gaussian. The combination of the energy differences for the 1^{st} and 2^{nd} 100ns is shown in figure 9. Although the distribution fits a Gaussian well, the energy differences show a large range: approximately 138 kcal/mol. When considering that it is the exponential of these energy differences that are calculated, 138 is extremely large. Nevertheless, by introducing a Gaussian to act as a smoothing function, the low-energy tail may be better described and provide convergence of the free energy.

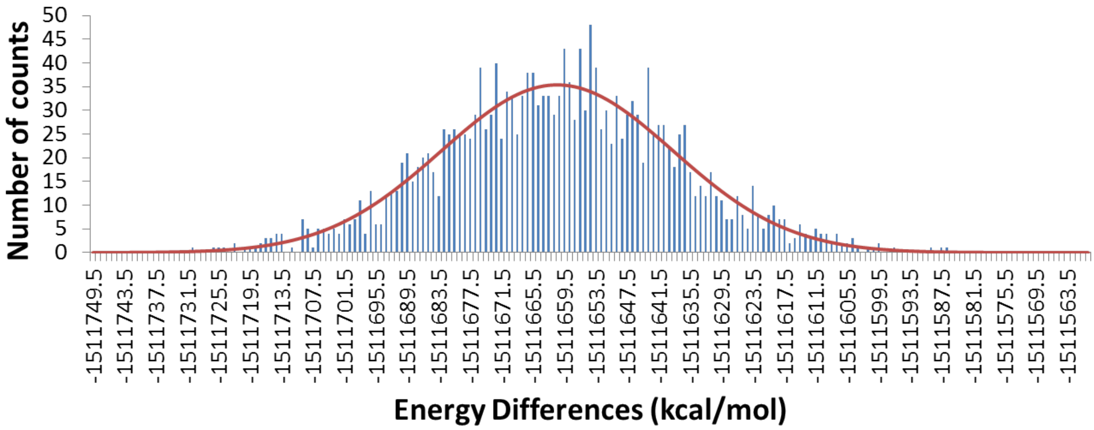

Figure 9: Energy differences for the full 200 ns of data for ethanol in 138 water

Using a Gaussian distribution to describe the energy differences for both the 1^{st} and 2^{nd} 100ns separately, then applying equation 2, the free energy can be found. By using these Gaussian, we approximate how the distribution would look if we could sample infinitely. To do this, the Gaussian is used as the probability density for ΔE, then the following can be used[3]:

$$\begin{aligned} \Delta G &= -k_B T \ln \int P(\Delta E) \exp\left[-\frac{\Delta E}{k_B T}\right] dE \\ &= -k_B T \left\langle -\frac{\Delta E}{k_B T} \right\rangle . \end{aligned} \tag{2}$$

Where $P(\Delta E)$ is the probability density of ΔE, k_B is the Boltzmann constant and T is the temperature.

In practice, however, the distribution was divided into small segments and the following equation was used:

$$\Delta G = -k_B T \ln \sum_{i=1}^{n_{Bars}} P(\Delta E_i) \exp\left[\frac{-\Delta E_i}{k_B T}\right]. \tag{3}$$

In order to calculate the required exponential of the energy difference for equation 3, Berg's [4] formula (equation !!!REF ME!!!) must be used, as before in section !!!REF ME!!!. In this case, however, by using the rules for dealing with logarithms, $\ln A$ will be the following,

$$\ln A = \ln(P(\Delta E_i)) - \frac{\Delta E_i}{k_B T}. \tag{4}$$

When using bin widths of 1 kcal/mol, where "bin" in this case refers to the small segments the Gaussian distribution is divided into, the results can be seen in table 9.

Table 9: Using Gaussian distributions for the 1^{st} and 2^{nd} 100 ns, then integrating over the probability distribution

1000 snapshots Bin width (kcal/mol)	1^{st} 100.0 ns Δ G	2^{nd} 100.0 ns Δ G	Difference ΔΔ G
1	-1512064.46	-1512070.75	6.29

Although the free energy is still not converged, the difference is a great deal smaller, down to 6.29 kcal/mol. This is a promising result, and the application of smoothing functions should be further investigated. For comparison, this method was applied directly to the histogram that was used to generate the Gaussians. The histograms were initially normalised so that the area underneath them was equal to 1, equation !!!REF ME!!!(BERG)was then applied using this new count. The results are shown in table 10.

Table 10: Using the histograms for the 1^{st} and 2^{nd} 100 ns

1000 snapshots Bin width (kcal/mol)	1^{st} 100.0 ns Δ G	2^{nd} 100.0 ns Δ G	Difference ΔΔ G
10	-1511729.23	-1511759.23	30.00
1	-1511723.55	-1511762.55	39.00
0.1	-1511722.84	-1511761.63	38.79

As expected without a smoothing function, the free energy does not converge. The funtional form of the Gaussian used as the smoothing function was the following,

$$P(\Delta E) = N \exp\left[-\alpha(E - E_0)^2\right]. \tag{5}$$

By integration of the following equation (equation 7) the analytical solution to the free energy can be obtained.

$$\begin{aligned}
\Delta G &= -k_B T \ln \left\langle \exp\left[-\frac{\Delta E}{k_B T}\right]\right\rangle && (6)\\
&= -k_B T \ln \frac{\int P(\Delta E)\exp\left[\frac{-\Delta E}{k_B T}\right] d\Delta E}{\int P(\Delta E) d\Delta E} && (7)\\
&= \frac{I_1}{I_2} && (8)
\end{aligned}$$

Solving I_2 is the simpler of the two integrals, as such it will be examined first.

$$I_2 = \int_{-\infty}^{\infty} \exp\left[-\alpha(\Delta E - \Delta E_0)^2\right] dE. \tag{9}$$

Setting $y = \sqrt{\alpha}(\Delta E - \Delta E_0)$,

$$\begin{aligned}
y &= \sqrt{\alpha}\Delta E - \sqrt{\alpha}E_0 && (10)\\
\frac{dy}{d\Delta E} &= \sqrt{\alpha} && (11)
\end{aligned}$$

$$d\Delta E = \frac{1}{\sqrt{\alpha}} dy \tag{12}$$

Substituting equation 12 into equation 9 and using the rule of integrating Gaussians, $\int_{-\infty}^{\infty} \exp[-x^2]dx = \sqrt{\pi}$, leads to,

$$I_2 = \frac{1}{\sqrt{\alpha}} \int_{-\infty}^{\infty} \exp[-y^2]dy = \sqrt{\frac{\pi}{\alpha}}. \tag{13}$$

Now, examining the numerator of equation 7,

$$\begin{aligned} I_1 &= \int_{-\infty}^{\infty} \exp\left[-\beta\Delta E\right] \exp\left[-\alpha(\Delta E - \Delta E_0)^2\right] d\Delta E & (14)\\ &= \int_{-\infty}^{\infty} \exp\left[-\beta\Delta E - \alpha(\Delta E - \Delta E_0)^2\right] d\Delta E & (15) \end{aligned}$$

Then, working on the exponent,

$$\begin{aligned} & -\beta\Delta E - \alpha(\Delta E - \Delta E_0)^2 & (16)\\ = & -\left[\beta\Delta E + \alpha\Delta E^2 + \alpha\Delta E_0^2 - 2\alpha\Delta E\Delta E_0\right] & (17)\\ = & -\left[\Delta E\left(\beta - 2\alpha\Delta E_0\right) + \alpha\Delta E^2 + \alpha\Delta E_0^2\right] & (18)\\ = & -\alpha\left[\Delta E\left(\frac{\beta}{\alpha} - 2\Delta E_0\right) + \Delta E^2 + \Delta E_0^2\right] & (19)\\ = & -\alpha\left[\Delta E^2 + 2\Delta E\left(\frac{\beta}{2\alpha} - \Delta E_0\right) + \left(\frac{\beta}{2\alpha} - \Delta E_0\right)^2 + \Delta E_0^2 - \left(\frac{\beta}{2\alpha} - \Delta E_0\right)^2\right] & (20)\\ = & -\alpha\left[\left(\Delta E + \left(\frac{\beta}{2\alpha} - \Delta E_0\right)\right)^2 + \Delta E_0 - \left(\frac{\beta}{2\alpha} - \Delta E_0\right)^2\right] & (21)\\ = & -\alpha\left[\left(\Delta E + \left(\frac{\beta}{2\alpha} - \Delta E_0\right)\right)^2 - \frac{\beta^2}{4\alpha^2} + \frac{\beta\Delta E_0}{\alpha}\right] & (22)\\ = & -\alpha\left(\Delta E + \left(\frac{\beta}{2\alpha} - \Delta E_0\right)\right)^2 - \frac{\beta^2}{4\alpha} + \beta\Delta E_0. & (23) \end{aligned}$$

Therefore,

$$\begin{aligned} I_1 &= \int_{-\infty}^{\infty} \exp\left[-\alpha(\Delta E - \Delta E_0)^2 - \beta\Delta E\right] d\Delta E & (24)\\ &= \int_{-\infty}^{\infty} \exp\left[-\alpha\left(\Delta E + \left(\frac{\beta}{2\alpha} - \Delta E_0\right)\right)^2 - \frac{\beta^2}{4\alpha} + \beta\Delta E_0\right] d\Delta E & (25)\\ &= \exp\left[-\frac{\beta^2}{4\alpha} + \beta\Delta E_0\right] \int_{\infty}^{\infty} \exp\left[-\alpha\left(\Delta E + \left(\frac{\beta}{2\alpha} - \Delta E_0\right)\right)^2\right] d\Delta E. & (26) \end{aligned}$$

Then,

$$x = \sqrt{\alpha}\left[\Delta E + \left(\frac{\beta}{2\alpha} - \Delta E_0\right)\right] \tag{27}$$

$$\frac{dx}{d\Delta E} = \sqrt{\alpha} \tag{28}$$
$$d\Delta E = \frac{1}{\sqrt{\alpha}}dx. \tag{29}$$

I_1 is then,

$$I_1 = \frac{1}{\sqrt{\alpha}}\exp\left[-\frac{\beta^2}{4\alpha}+\beta\Delta E_0\right]\int_{-\infty}^{\infty}\exp\left[-x^2\right]dx \tag{30}$$
$$= \sqrt{\frac{\pi}{\alpha}}\exp\left[-\frac{\beta^2}{4\alpha}+\beta\Delta E_0\right]. \tag{31}$$

Combining I_1 and I_2 gives,

$$\frac{I_1}{I_2} = \langle\exp\left[-\beta\Delta E\right]\rangle \tag{32}$$
$$= \exp\left[-\frac{\beta^2}{4\alpha}+\beta\Delta E_0\right]. \tag{33}$$

Inserting this back into the Zwanzig equation,

$$\Delta G = -\beta^{-1}\ln\langle\exp\left[-\beta\Delta E\right]\rangle \tag{34}$$
$$= -\beta^{-1}\ln\frac{I_1}{I_2} \tag{35}$$
$$= -\beta^{-1}\ln\left(\exp\left[-\frac{\beta^2}{4\alpha}+\beta E_0\right]\right) \tag{36}$$
$$= -\beta^{-1}\left(\frac{\beta^2}{4\alpha}-\beta E_0\right) \tag{37}$$
$$= E_0-\frac{\beta}{4\alpha} \tag{38}$$

It has been shown that the free energy can then be found directly from the exponents of the Gaussian,

$$\Delta G = E_0-\frac{1}{4\alpha k_B T}. \tag{39}$$

The exponents fitted when using the full histogram are shown in table 11 and the free energies calculated when using these exponents within equation 39 are shown in table 12. Along with these free energies, are the resulting free energies when the sample size for the histogram is reduced and the Gaussian re-fitted. When lowering the sample size, the snapshots selected were spread evenly throughout the simulation.

Table 11: Exponents used for the Gaussian distributions when using the full histograms for the 1^{st} and 2^{nd} 100 ns

coefficient	1^{st} 100.0 ns	2^{nd} 100.0 ns	Difference
α	0.0010276	0.0010126	0.000015
E_0	-1511661.01	-1511660.96	-0.06

Table 12: Free energies obtained when using a Gaussian probability density along with equation 39 for the 1^{st} and 2^{nd} 100 ns

	1^{st} 100.0 ns Δ G (kcal/mol)	2^{nd} 100.0 ns Δ G (kcal/mol)	Difference $\Delta\Delta$ G (kcal/mol)
1000	-1512069.09	-1512075.07	5.98
500	-1512087.09	-1512076.93	-10.16
100	-1512232.73	-1512010.76	-221.97

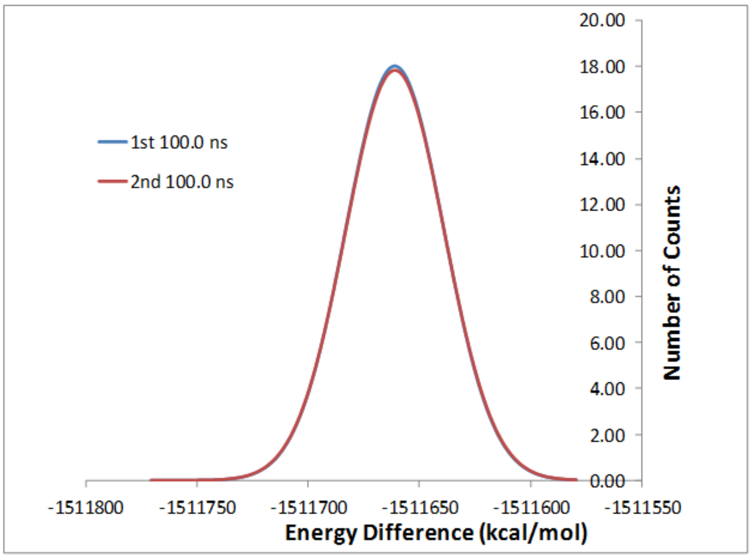

Figure 10: The Gaussian distributions for the 1^{st} and 2^{nd} 100 ns

The difference in free energies is surprising when considering the small difference between the exponents of the Gaussian distribution. Figure 10 shows the two distributions overlapping to give an indication of how similar these distributions are. These results led to a simple numerical test to find how sensitive these exponents are. First, $\Delta\Delta$ G can be calculated by,

$$\begin{aligned}
\Delta\Delta G &= \left(E_0^{(1)} - \frac{1}{4\alpha_1 k_B T}\right) - \left(E_0^{(2)} - \frac{1}{4\alpha_2 k_B T}\right) && (40)\\
&= \Delta E_0 - \frac{1}{4\alpha_1 k_B T} + \frac{1}{4\alpha_2 k_B T} && (41)\\
&= \Delta E_0 - \frac{1}{\alpha_1} + \frac{1}{\alpha_2}. && (42)
\end{aligned}$$

By setting ΔE_0 to 0, the α exponents can be tested. The aforementioned numerical test then involved setting α_1 to values ranging from 0.001 to 0.999 in incremental steps of 0.001 and α_2 was calculated to be $\alpha_1 + 1\%$ and $\alpha_1 - 1\%$. The two $\Delta\Delta G$ values obtained were then compared and are shown in figure 11.

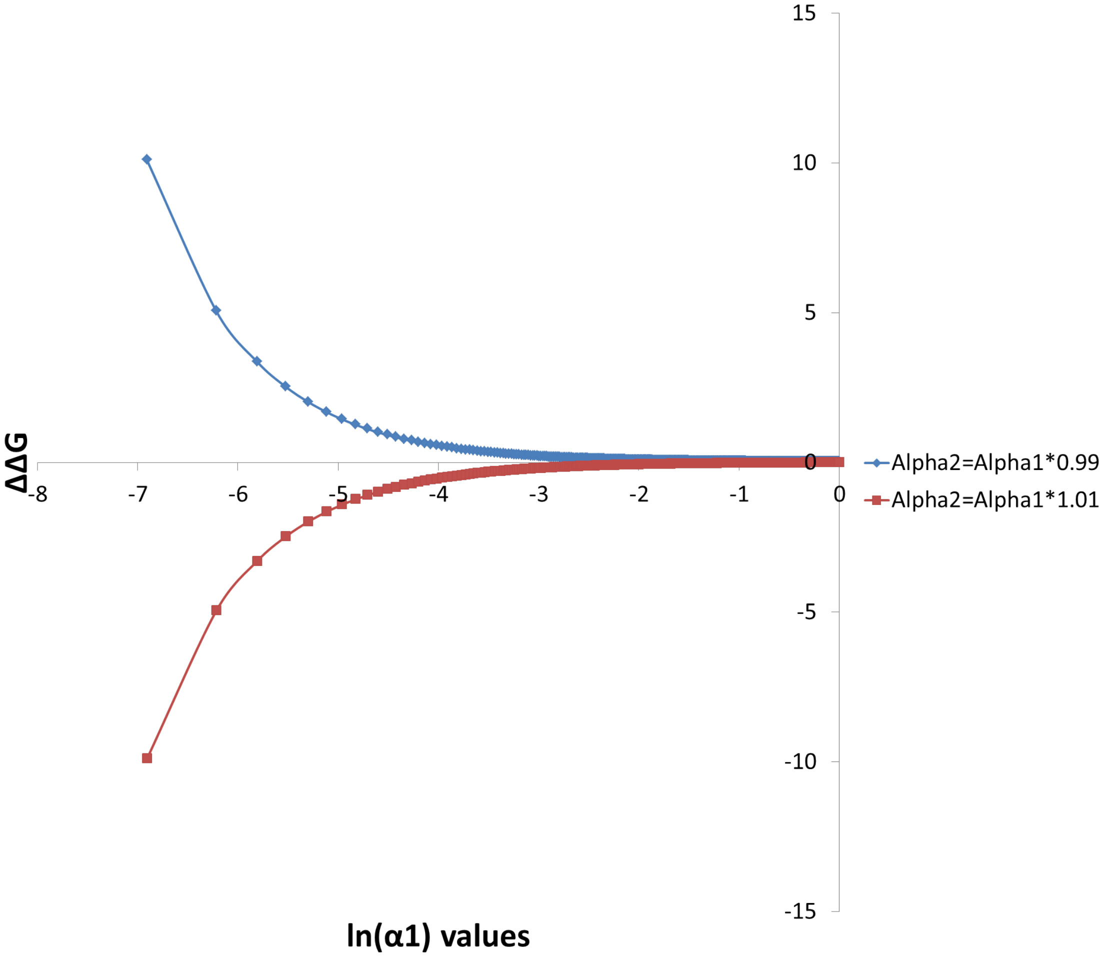

Figure 11: The sensitivity of the α exponents used in equation 39

It is clear that when α is very small there is little room for any error. However, the relative free energy for the system in table 12 shows a reduction from applying the Zwanzig equation directly. The reason for this can be found when considering that the exponential average is not calculated, so the low-energy tail of the distribution is not given unfair weighting. To avoid this exponential averaging issue, the cumulant expansion of the free energy was considered. This is simply the expansion of the Zwanzig equation (equation 2) in a Taylor series [5], the derivation follows.

Firstly we must specify the functional form that will be used,

$$\begin{aligned} \Delta G &= -\frac{1}{\beta}\ln\langle\exp\left[-\beta\Delta E\right]\rangle && (43)\\ &= -\frac{1}{\beta}\ln\left\langle\sum_{l=0}^{\infty}\frac{(-\beta\Delta E)^l}{l!}\right\rangle && (44)\\ &= -\frac{1}{\beta}\ln\left(1+\sum_{l=1}^{\infty}\frac{(-\beta)^l}{l!}\left\langle\Delta E^l\right\rangle_0\right). && (45) \end{aligned}$$

The step from equation 44 to 45 is performed as $\ln(1+x)$ can be expanded in a Maclaurin series (expansion of a function using a Taylor series at 0).

$$f(x) = \sum_{n=0}^{\infty} f^{(n)}(a)\frac{(x-a)^n}{n!} \tag{46}$$

$$
\begin{aligned}
f(a) &= \ln(1+a) \\
f'(a) &= (1+a)^{-1} \\
f''(a) &= -(1+a)^{-2} \\
f'''(a) &= 2(1+a)^{-3} \\
f''''(a) &= -6(1+a)^{-4} \\
&\dots \\
a &= 0 \\
f(x) &= \ln(1+0) + (1+0)^{-1}\frac{x}{1!} + -(1+0)^{-2}\frac{x^2}{2!} + 2(1+0)^{-3}\frac{x^3}{3!} - 6(1+0)^{-4}\frac{x^4}{4!} \qquad (47) \\
&= 0 + x - \frac{x^2}{2} + \frac{2x^3}{6} - \frac{6x^4}{24} \qquad (48) \\
&= 0 + x - \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} \qquad (49) \\
&= \sum_{k=1}^{\infty}(-1)^{n-1}\frac{x^n}{n} \qquad (50)
\end{aligned}
$$

Combining equation 45 and 50,

$$
\Delta G = -\frac{1}{\beta}\sum_{k=1}^{\infty}(-1)^{k-1}\frac{1}{k}\left(\sum_{l=1}^{\infty}\frac{(-\beta)^l}{l!}\left\langle \Delta E_1^l \right\rangle\right) \qquad (51)
$$

Equating 51 to the cumulant generating function, equation 52, and cancelling a factor of $1/\beta$,

$$
\Delta G = \sum_{k=1}^{\infty}\frac{(-\beta)^{k-1}}{k!}\omega_k. \qquad (52)
$$

Extracting powers of β^1

$$
\begin{aligned}
-\beta\omega_1 &= -\beta\langle \Delta E\rangle_0 \\
\omega_1 &= \langle \Delta E\rangle_0. \qquad (53)
\end{aligned}
$$

Next, extracting powers of $(\beta)^2$,

$$
\begin{aligned}
\frac{(-\beta)^2}{2}\omega_2 &= \left(\frac{(-\beta^2)}{2}\langle \Delta E^2\rangle_0\right) - \frac{1}{2}\left(\frac{(-\beta)}{1}\langle \Delta E\rangle_0\right)^2 \qquad (54) \\
&= \frac{(-\beta)^2}{2}\langle \Delta E^2\rangle_0 - \frac{(-\beta)^2}{2}\langle \Delta E\rangle_0^2 \qquad (55) \\
\omega_2 &= \langle \Delta E^2\rangle_0 - \langle \Delta E\rangle_0^2 \qquad (56) \\
\langle \Delta E^2\rangle_0 - \langle \Delta E\rangle_0^2 &= \langle(\Delta E - \langle \Delta E\rangle_0)^2\rangle_0 \qquad (57) \\
&= \langle \Delta E^2 + \langle \Delta E\rangle^2 - 2\Delta E\langle \Delta E\rangle_0\rangle_0 \qquad (58) \\
&= \langle \Delta E^2\rangle_0 + \langle \Delta E\rangle_0^2 - 2\langle \Delta E\rangle_0^2 \qquad (59) \\
&= \langle \Delta E^2\rangle_0 - \langle \Delta E\rangle_0^2. \qquad (60)
\end{aligned}
$$

Extracting β^3,

$$\begin{aligned}
\frac{-\beta^3}{6}\omega_3 &= -\beta\langle\Delta E\rangle_0 + \frac{(-\beta)^2}{2}\langle\Delta E^2\rangle_0 + \frac{(-\beta)^3}{6}\langle\Delta E^3\rangle_0 \\
&- \frac{1}{2}\left(-\beta\langle\Delta E\rangle_0 + \frac{(-\beta)^2}{2}\langle\Delta E^2\rangle_0 + \frac{(-\beta)^3}{6}\langle\Delta E^3\rangle_0\right)^2 \\
&+ \frac{1}{3}\left(-\beta\langle\Delta E\rangle_0 + \frac{(-\beta)^2}{2}\langle\Delta E^2\rangle_0 + \frac{(-\beta)^3}{6}\langle\Delta E^3\rangle_0\right)^3 \quad (61)\\
&= \frac{(-\beta)^3}{6}\langle\Delta E^3\rangle_0 - \frac{(-\beta)^3}{4}\langle\Delta E^2\rangle_0\langle\Delta E\rangle_0 + \frac{(-\beta)^3}{3}\langle\Delta E\rangle_0^3 \quad (62)\\
\omega_3 &= \langle\Delta E^3\rangle_0 - 3\langle\Delta E^2\rangle_0\langle\Delta E\rangle_0 + 2\langle\Delta E\rangle_0^3. \quad (63)
\end{aligned}$$

Additional powers of β can be extracted in the same manner, however the higher the power, the more complicated the derivation. When calculating the free energy using the cumulant expansion and applying a Gaussian distribution as a smoothing function, the terms naturally truncate after the first 2 cumulants [6]. This should not come as a surprise, examination of the first 2 cumulants, $\langle\Delta E\rangle_0$ and $\langle\Delta E^2\rangle_0 - \langle\Delta E\rangle_0^2$ are equal to the average energy and the variance respectively. With these two pieces of information a Gaussian distribution can be plotted. The third cumulant is equal to the slant of the data, as Gaussian has no slant, this is equal to 0. Additionally, what should be noticed is that, when applied to a Gaussian, this approach is identical to the analytical approach previously derived,

$$\begin{aligned}
\Delta E_0 &= \langle\Delta E\rangle_0 \quad (64)\\
\sigma^2 &= \frac{1}{2\alpha} \quad (65)\\
\sigma^2 &= \langle\Delta E^2\rangle_0 - \langle\Delta E\rangle_0^2 \quad (66)\\
\Delta G &= \langle\Delta E\rangle_0 - \frac{\beta}{2}(\langle\Delta E^2\rangle_0 - \langle\Delta E\rangle_0^2) \quad (67)\\
&= \Delta E_0 - \frac{\beta}{2}\left(\frac{1}{2\alpha}\right). \quad (68)
\end{aligned}$$

Applying this method directly to the histogram without the use of a smoothing function produced the following results shown in table 13.

Table 13: Applying the cumulant expansion to free energy directly to the histogram, all values in kcal/mol

Cumulant truncation	1^{st} 100.0 ns	2^{nd} 100.0 ns	Difference
First	-1511660.60	-1511660.94	0.34
Second	-1511245.27	-1511233.63	-11.64
Third	-1511725.47	-1512194.03	468.56

Truncation after the first cumulant shows good convergence between the 2 runs, however, this is just the average energy difference and cannot be considered a full solution the free energy. Truncating at the second cumulant proves that the free energy convergence is strongly negatively affected by the variance, i.e. the distribution is spread over a large range of data.

In an attempt to lower the affect of the variance on the free energy a method, similar to that of bootstrapping, of extracting random samples was used. The process follows,

1. Extract 500 snapshots
2. Fit a Gaussian distribution
3. Repeat 100 times
4. Average the exponents
5. Apply the analytical equation for the free energy

Table 14: Average exponents after a "random samples" approach

Coefficient	1^{st} 100.0 ns	2^{nd} 100.0 ns	Difference
α	0.0010383	0.0010194	0.0000189
ΔE_0	-1511661.49	-1511661.48	-0.01

Table 15: Average exponents after a "random samples" approach

	1^{st} 100.0 ns	2^{nd} 100.0 ns	Difference
Snapshots	ΔG (kcal/mol)	ΔG (kcal/mol)	$\Delta\Delta G$ (kcal/mol)
1000			

BOOTSTRAPPING SIMILARITY METHOD

HISTOGRAM CUTOFFS

GAUSSIAN CUTOFF

MULTISTATE GAUSSIANS

CONCLUSIONS

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